

CariSurg MedTech Pathways Research Project

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Statement of the Problem

Mercer General Hospital's ED faces risk of patient under-triage during peak periods, where nurse workload and time pressure can cause high-risk cases to be missed. A 12-week pilot will test an explainable AI support tool that uses structured triage data and free-text presenting complaints to flag high-risk cases for nurse review, displaying a visible reason for each alert. Nurses retain full accountability for all final escalation decisions.

Previous Work

A growing body of literature has examined artificial intelligence in emergency department triage. The ten studies below form the evidence base for this proposal.

Tyler et al. (2024)

A scoping review found that AI and machine-learning tools show promise for acuity prediction, resource allocation, and earlier recognition of urgent cases, while noting that most evidence remains retrospective.

Da'Costa et al. (2025)

This review highlights potential improvements in triage speed and consistency, but identifies bias, data quality, explainability, privacy, and clinician trust as major implementation concerns.

Araouchi & Adda (2024)

TriageIntelli used multimodal triage data and achieved strong KTAS classification performance, but its single-dataset design limits generalisability.

Chen et al. (2024)

Machine-learning and NLP models using structured triage data and free-text notes improved disposition prediction, though longer-term outcomes and resource use were not fully addressed.

Ottenhoff et al. (2023)

Machine learning may support more consistent triage and identify patients at risk of adverse outcomes, but live clinician trust and workflow impact remain unresolved.

Townsend et al. (2023)

Practitioners saw value in AI support but raised concerns about accountability, transparency, bias, and over-reliance. **Hinson et al. (2024)**

A multisite AI-informed decision-support implementation improved triage equity across demographic groups, providing a real-world precedent for fairness-focused deployment.

De Freitas et al. (2020)

This Caribbean ED patient-flow study provides the regional context for workflow mapping, including CTAS usage, resource constraints, and high attendance volumes.

Cheng et al. (2024)

Implementation facilitators include staff education, efficient workflow integration, and external validation; barriers include clinician resistance and poor change management.

Zayas-Caban et al. (2023)

This paper identifies data quality, EHR integration, clinician trust, and regulatory readiness as key adoption barriers in emergency care.

Gaps in the Literature

- Many studies report retrospective model performance, but fewer evaluate whether AI improves live nurse triage decisions in real ED workflows.
- Accuracy metrics are common, while alert burden, explainability, demographic fairness, and clinician actionability are less consistently measured.
- No identified study evaluates AI triage performance in a Caribbean ED context, where local workflow constraints may affect model performance and adoption.

Proposed Solution

This pilot proposes an explainable AI support tool for emergency department triage at Mercer General Hospital. It will process structured triage inputs, including vital signs, chief complaints, and acuity scores, alongside free-text presenting complaint notes to identify patients at elevated risk of under-triage.

The model will combine gradient boosting and natural language processing to flag high-risk patients for nurse review. Each alert will display a human-readable reason, such as elevated heart rate combined with complaint keywords associated with sepsis, so nurses can evaluate or override the recommendation with full clinical accountability.

The 12-week pilot will measure sensitivity, specificity, alert burden, fairness across demographic groups, and effect on escalation decision time.

ED Triage Workflow & AI Integration Points

Workflow Step	AI / System Note
Patient arrives	
↓	
Door screen Decision: Level 1?	AI candidate: queue wait and deterioration risk
↓	
Registration Admin record created	
↓	
Vitals captured Paper card or EHR	AI candidate: vital-sign anomaly check
↓	
Triage nurse assessment Chief complaint + CTAS	CORE AI support: complaint NLP + risk score shown beside CTAS; nurse decides
↓	
Clinical routing Immediate / Urgent / Minor Care	
↓	
Physician assessment	AI candidate: alert if CTAS conflicts with vitals or complaint evidence
↓	
Disposition Admit / Observe / Discharge	AI candidate: post-discharge deterioration risk flag
Constraints	EHR data lag; nurse cognitive load and alert fatigue; free-text documentation variability

(Note that I encountered errors importing the svg into the document so I just created a demo table, wanted to make this work as visually pleasing as possible, please refer to the diagram that would be attached in github, alongside this deliverable)

The table below describes the current ED triage workflow from door to disposition. Wording has been tightened so each row clearly identifies the workflow stage and the specific AI function without fragmented table text.

Step	Stage	AI Integration Point
1	Patient arrival and door screen	Predict queue wait time and pre-flag patients at risk of deterioration before formal assessment.
2	Vitals capture: BP, HR, SpO2, Temp	Detects abnormal vital-sign combinations before nurse assessment.
3	Triage nurse assessment: chief complaint + CTAS	CORE: Run NLP on free-text notes and show a gradient-boosting risk score beside the CTAS level.
4	Physician assessment	Alert if the assigned CTAS level conflicts with vital signs or complaint keyword patterns.
5	Disposition decision	Flag patients likely to deteriorate after discharge based on the full visit profile.

Constraints & Stakeholders

Workflow Constraints

Constraint 1: EHR Data Lag. Vitals are frequently recorded on paper triage cards before manual EHR entry, introducing a 5-30 minute lag. The tool must either interface with bedside monitoring equipment or tolerate delayed data with explicit uncertainty flagging.

Constraint 2: Nurse Cognitive Load and Alert Fatigue. Triage nurses manage simultaneous assessments, family communication, and documentation. Alerts must be rare, high-confidence, and readable at a glance.

Constraint 3: Free-Text Documentation Variability. The richest NLP input is also the most variable. The model must be calibrated on locally sourced ED text samples before deployment.

Clinical Stakeholders

Stakeholder	Role	Top Concern
Triage Nurse (Sr. Patrice Alleyne)	First clinical contact; assigns CTAS level.	Will this slow me down or add steps during a busy shift?
ED Consultant Physician	Final clinical authority; makes disposition decisions.	Is the AI accountable, auditable, and governed clinically?
Clinical IT Lead (Martina Griffith)	Oversees integration, privacy, and audit requirements.	Does this meet data governance, privacy, and security standards?
Patient / Patient Advocate	Recipient of the triage decision.	Will I be treated fairly regardless of background, language, or insurance status?
Ward Porter / Healthcare Assistant	Transfers patients between ED areas.	Will alert-driven escalations increase workload unpredictably?

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